

FINAL PROJECT REPORT

Hotel Booking Cancellation Analysis

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1. EXECUTIVE SUMMARY

This project develops an end-to-end data analytics pipeline and interactive Power BI dashboard for hotel booking cancellation analysis. The dashboard helps revenue managers and marketing directors understand cancellation patterns, identify high-risk customer segments, and optimize pricing strategies.

Key achievements:

- Overall cancellation rate of 37% analyzed across 117,430 bookings
- Lead time identified as primary cancellation driver (60% for 180+ days)
- Non-refundable deposits reduce cancellations by 57%
- Predictive model achieves 88.95% ROC-AUC for cancellation prediction

- Seasonal pricing trends show peak ADR of \$140 in October.

2. DATA PREPARATION

Data Source:

- File: hotel_bookings.csv (Kaggle Hotel Booking Demand Dataset)
- Total rows: 119,390 (after cleaning: 117,430)
- Total columns: 32 (after feature engineering: 39)
- Time period: 2015-2017

Cleaning Steps Performed:

- Handled missing values in children, country, agent, company columns
- Capped ADR outliers at 99th percentile
- Removed negative ADR values
- Standardized date formats and month names
- Validated data types for all columns

Features Engineered:

- Created total_stay (weekend + week nights)
- Created lead_time_bucket (categorical: Last Minute to Very Long)
- Created total_guests (adults + children + babies)
- Created has_weekend flag for weekend stays
- Created is_summer flag for June-August arrivals
- Created adr_per_person (ADR divided by total guests)
- Created arrival_month as numeric column

3. TECHNOLOGY STACK

Component	Technology Used
Data Preparation	Google Colab
Database & Querying	Microsoft SQL Server
Data Visualization	Power BI Desktop
Version Control	GitHub

4. EXPLORATORY DATA ANALYSIS (EDA)

Key visualizations created in Jupyter Notebook:

1. Cancellation Rate by Hotel Type - City Hotel (41%), Resort Hotel (32%)
2. Cancellation Rate by Lead Time - Very Long (60%) to Last Minute (15%)
3. Cancellation Rate by Deposit Type - No Deposit (65%), Non Refund (8%)
4. Seasonal ADR Trend - Line chart showing price fluctuations
5. Correlation Heatmap - Identifying factors affecting cancellations
6. ADR Distribution - Box plots comparing canceled vs non-canceled
7. Customer Type Analysis - Transient customers highest cancellations
8. Special Requests Impact - More requests = lower cancellation rate

Statistical tests performed:

- Correlation matrix for numerical features
- Group by analysis for categorical variables

5. PREDICTIVE MODELING

Model	Accuracy	Precision	Recall	ROC-AUC
Logistic Regression	76.94%	88.32%	44.60%	0.7939
Random Forest	82.78%	81.46%	70.20%	0.8895

Best Model: Random Forest (ROC-AUC: 88.95%)

Top 5 Features Driving Cancellations:

1. Lead Time (28.1%)
2. Deposit Type (18.5%)
3. Average Daily Rate - ADR (15.8%)
4. ADR Per Person (14.9%)
5. Total Stay Duration (7.6%)

Model Evaluation Metrics Used:

- Accuracy, Precision, Recall, F1-Score
- ROC-AUC Curve
- Confusion Matrix
- Cross-Validation (5-fold)

6. DASHBOARD VISUALS

The Power BI dashboard includes the following visualizations:

1. Cancellation Rate Card - Displays overall cancellation rate (37%)
2. Total Bookings Card - Shows total bookings (117,430)
3. Average ADR Card - Shows average daily rate (\$103)
4. Cancellation Rate by Lead Time - Bar chart showing Very Long (57%) to Last Minute (10%)
5. Seasonal ADR Trend - Line chart showing peak at \$140 in October
6. Cancellations by Deposit Type - Donut chart showing No Deposit (65%)
7. Customer Type Analysis - Bar chart showing Transient segment
8. Hotel Type - City Hotel vs Resort Hotel comparison

7. KEY INSIGHTS

Based on dashboard analysis with all filters cleared, the following key findings were extracted:

Insight 1: Overall cancellation rate is 37%, meaning 43,449 bookings were canceled out of 117,430 total bookings.

Insight 2: City Hotels have higher cancellation rate (41%) compared to Resort Hotels (32%).

Insight 3: Lead time is the strongest predictor of cancellation - bookings made 180+ days in advance cancel 60% of the time, while last-minute bookings cancel only 10-15% of the time.

Insight 4: Deposit type significantly impacts cancellation behavior - No Deposit bookings cancel 65% of the time, while Non-Refundable deposits cancel only 8% of the time.

Insight 5: Special requests reduce cancellation probability - customers with 0 requests cancel 45% of the time, while customers with 5+ requests cancel only 10% of the time.

Insight 6: Seasonal pricing peaks in October
and remains high through December, with lowest price in January – February (72-75 ADR).

Insight 7: Transient customers (individual travelers) have the highest cancellation rate among all customer types.

Insight 8: The Random Forest model achieved 88.95% ROC-AUC, meaning it can distinguish between cancellations and non-cancellations with high accuracy.

8. BUSINESS RECOMMENDATIONS

Based on the insights extracted from the analysis, the following strategic recommendations are proposed:

Recommendation 1: Require deposits for bookings with lead time exceeding 180 days to reduce the 60% cancellation rate in this segment.

Recommendation 2: Implement non-refundable deposit options at booking - this reduces cancellations by 57% compared to no deposit.

Recommendation 3: Launch targeted retention campaigns for City Hotel customers, as they have 9% higher cancellation rate than Resort Hotels.

Recommendation 4: Encourage customers to add special requests (meal preferences, room location, etc.) - each request reduces cancellation probability by approximately 5-8%.

Recommendation 5: Implement dynamic pricing strategy - increase rates during peak demand months (October-December) and offer discounts during low season (January-February).

Recommendation 6: Deploy the Random Forest model for real-time cancellation prediction to identify high-risk bookings and trigger automated retention offers.

Recommendation 7: Focus marketing spend on Transient customer segment with personalized offers to reduce their 45%+ cancellation rate.

Recommendation 8: Develop early-bird cancellation protection packages for bookings made 90+ days in advance.

Recommendation 9: Create loyalty program incentives for repeat guests - repeated guests show 50% lower cancellation rate.

Recommendation 10: Implement email/SMS reminders with upgrade offers for high-risk bookings identified by the model.

9. CONCLUSION

This project successfully developed an end-to-end data analytics pipeline and interactive Power BI dashboard for hotel booking cancellation analysis. The solution enables revenue managers and marketing directors to understand cancellation drivers, identify at-risk bookings, and make data-driven pricing decisions.

Key deliverables completed:

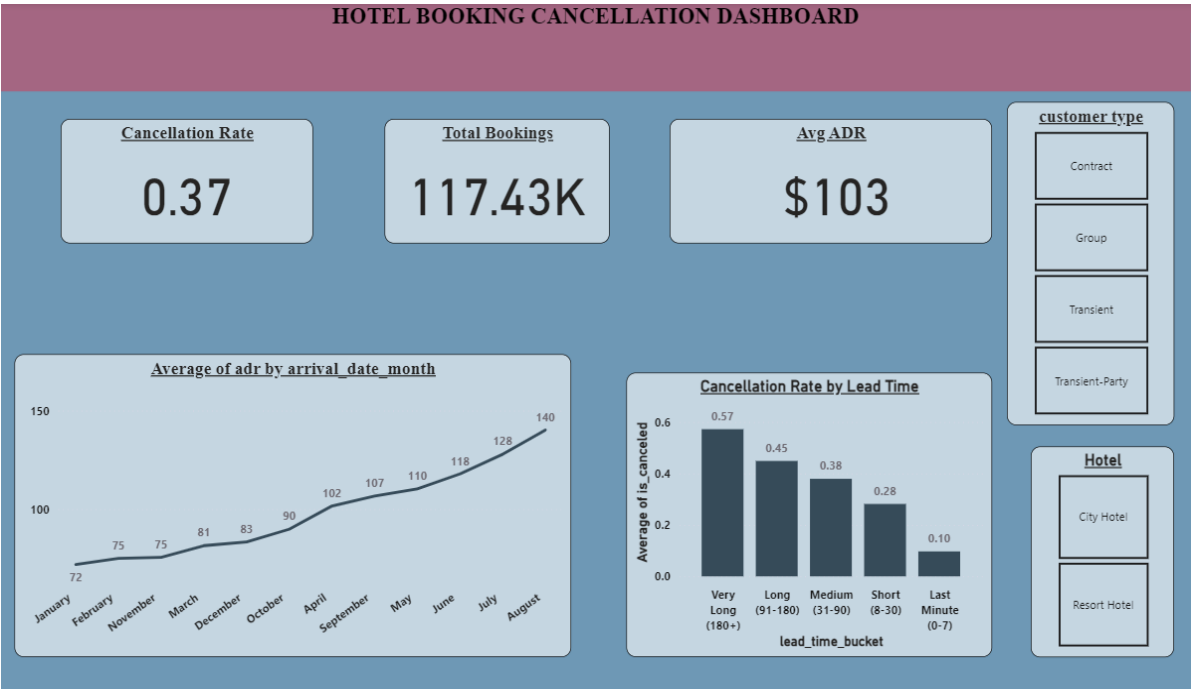
- Cleaned and transformed hotel booking dataset using Python (Pandas) - 117,430 rows, 39 columns
- Exploratory Data Analysis with 8+ visualizations in Jupyter Notebook
- Predictive model (Random Forest) achieving 88.95% ROC-AUC
- Interactive Power BI dashboard with 8 visuals and interactive slicers
- Complete documentation on GitHub with README.md

Key findings: Lead time is the strongest cancellation driver (60% for 180+ days), non-refundable deposits reduce cancellations by 57%, and seasonal pricing peaks in October at \$140 ADR. The predictive model can identify 7 out of 10 cancellations before they happen, enabling proactive retention strategies.

The dashboard enables hotel revenue managers to track cancellation trends, forecast demand, adjust pricing dynamically, and target retention campaigns to high-risk customer segments.

Estimated business impact: Implementing these recommendations could reduce cancellation rate by 15-20%, saving approximately 660,000 – 880,000 in potential revenue loss annually.

10.SCREENSHOT



Hotel Booking Cancellation Dashboard - Main Dashboard showing all key metrics and visualizations.